Uniform Boundedness Principle

An End-to-End Meta-Math Walkthrough

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Abstract

Convert bounds depending on the point into one common bound on a ball, then subtract its center and rescale. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

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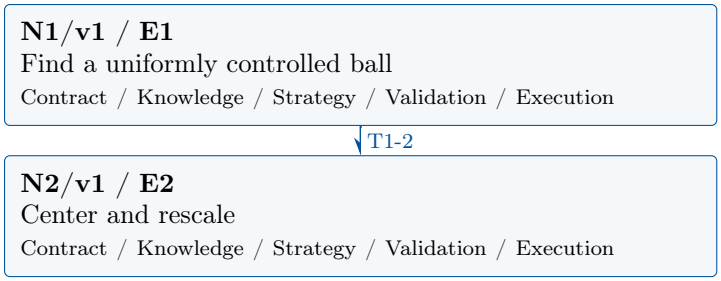
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let X be a Banach space, Y a normed space, and $\mathcal{T}$ a family of bounded linear maps $X \to Y$. If for every $x \in X$ the number $\sup_{T\in\mathcal{T}} \|Tx\|$ is finite, prove $\sup_{T\in\mathcal{T}} \|T\| < \infty$. The family need not be countable, and Y need not be complete.*

A. Foundations and permitted imports

Import the Baire category theorem: a nonempty complete metric space cannot be a countable union of closed sets with empty interior. Use continuity of bounded operators, the triangle inequality, and the operator norm. The empty family satisfies the boundedness conclusion vacuously; treat the nonempty case below.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. There are N, x_0, and $r > 0$ with $B(x_0, r) \subset F_N$.

Receiving activity. N2.1

Compatibility. E1/v1 and linearity of every T. Check: The ball is not necessarily centered at zero, so its center must be subtracted before scaling.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Find a uniformly controlled ball

Object and role. Apply Baire to the countable collection of level sets, even when the operator family is uncountable. **Local obligation:** There are N, x_0, and $r > 0$ with $B(x_0, r) \subset F_N$.

Inputs. $F_n = \{x : \sup_{T\in\mathcal{T}} \|Tx\| \le n\}$ for integers $n \ge 1$.

C — Contract

Publish E1/v1 only after establishing this obligation: There are N, x_0, and $r > 0$ with $B(x_0, r) \subset F_N$.

K — Knowledge

$F_n = \{x : \sup_{T\in\mathcal{T}} \|Tx\| \le n\}$ for integers $n \ge 1$. Retain the stated assumptions and import versions.

S — Strategy

Apply Baire to the countable collection of level sets, even when the operator family is uncountable.

V — Validation

Each F_n is an arbitrary intersection of closed inverse images; countability is needed for the cover by F_n, not for $\mathcal{T}$.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 For each T, the set $\{x : \|Tx\| \le n\}$ is closed by continuity, so their intersection F_n is closed. Pointwise boundedness says $X = \bigcup_{n\ge1} F_n$.

N1.2 The Banach space X is a nonempty complete metric space. Baire therefore gives an F_N with nonempty interior, which contains some open ball $B(x_0, r)$. In particular $x_0 \in F_N$.

Export E1/v1. There are N, x_0, and $r > 0$ with $B(x_0, r) \subset F_N$.

Acceptance gate. Each F_n is an arbitrary intersection of closed inverse images; countability is needed for the cover by F_n, not for $\mathcal{T}$.

Targeted repair. If Baire is applied to the operator index set, rebuild the countable level-set cover of the complete space X. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Center and rescale

Object and role. Use a difference of two points in the controlled ball to bound the unit ball.
Local obligation: For every $T \in \mathcal{T}$, $\|T\| \le 4N/r$.

Inputs. E1/v1 and linearity of every T. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: For every $T \in \mathcal{T}$, $\|T\| \le 4N/r$.

K — Knowledge

E1/v1 and linearity of every T. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Use a difference of two points in the controlled ball to bound the unit ball.

V — Validation

The ball is not necessarily centered at zero, so its center must be subtracted before scaling.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 For $\|h\| < r$, both $x_0 + h$ and x_0 belong to F_N, so $\|Th\| \le \|T(x_0 + h)\| + \|Tx_0\| \le 2N$ for every T.

N2.2 For $\|u\| \le 1$, choose $h = (r/2)u$. Then $\|Tu\| = (2/r)\|Th\| \le 4N/r$. Taking the supremum over the unit ball and then the family proves the theorem. Completeness of Y was never used.

Export E2/v1. For every $T \in \mathcal{T}$, $\|T\| \le 4N/r$.

Acceptance gate. The ball is not necessarily centered at zero, so its center must be subtracted before scaling.

Targeted repair. If a candidate rescales $x_0 + h$ directly, replace it by the difference h first; retain the factor two from the triangle inequality. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Find a uniformly controlled ball

1. **N1.1.** For each T, the set $\{x : \|Tx\| \le n\}$ is closed by continuity, so their intersection F_n is closed. Pointwise boundedness says $X = \bigcup_{n\ge 1} F_n$.
2. **N1.2.** The Banach space X is a nonempty complete metric space. Baire therefore gives an F_N with nonempty interior, which contains some open ball $B(x_0, r)$. In particular $x_0 \in F_N$.

B. N2 — Center and rescale

1. **N2.1.** For $\|h\| < r$, both x_0+h and x_0 belong to F_N, so $\|Th\| \le \|T(x_0+h)\|+\|Tx_0\| \le 2N$ for every T.
2. **N2.2.** For $\|u\| \le 1$, choose $h = (r/2)u$. Then $\|Tu\| = (2/r)\|Th\| \le 4N/r$. Taking the supremum over the unit ball and then the family proves the theorem. Completeness of Y was never used.

□

V. WORKED ORCHESTRATION

On ℓ^2, take $T_n x = x_n$. For each x, $\sup_n |x_n| \le \|x\|_2$. Testing the standard basis gives $\|T_n\| = 1$, so the pointwise and uniform bounds agree. The theorem also applies when a direct formula for the common norm bound is unavailable.

A. Boundary and hypothesis audit

On the incomplete normed space c_{00} with the sup norm, set $T_n x = n x_n$. Each T_n is bounded with norm n. For each finitely supported x, only finitely many $T_n x$ are nonzero, so pointwise boundedness holds while uniform boundedness fails.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected generalization (N1.2): the theorem works for every normed domain. **Validation:** c_{00} and $T_n x = n x_n$ refute it. **Diagnosis:** the Baire import requires completeness.

Repair: retain the original Banach-domain contract and apply Baire exactly there; extending the target to arbitrary normed spaces would require new assumptions. N2 may consume only the ball export proved under completeness.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Combine this with pointwise convergence of bounded operators to control their norms; convergence in operator norm still requires additional hypotheses.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. MIT OpenCourseWare, Introduction to Functional Analysis: lecture notes. See the lectures on integration, Hilbert spaces, Baire category, and Hahn–Banach.